

APWG eCrime eXchange: Data Exchange Intervention Against Cybercrime



*A Global Response to Cybercrime via Precision Data
Logistics and Rigorously Curated Data Exchange
Policies Since 2004*

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Cybercrime Response Data Logistics as Emerging Discipline

- Cybercrime interveners need a curated clearinghouse for machine event data, rigorously maintained for categorizational fidelity and report reliability – that is if ecosystem-level responses to cybercrime are ever to cohere into public-health models of intervention
- APWG's role, **since 2003**, has been to lead the global dialog for orchestrating indicator machine event data into *programmatically recruitable data corpora* for automating responses to the most common cybercrimes
- APWG's eCrime eXchange and its undergirding policies are products of that dialog with private and public sectors over the decades since APWG's foundation

Data Exchange Drives Routinized Responses to Common Risk Events

- APWG considers the public health model of intervention useful to emulate for suppression of cyber risks *at global scale*
- APWG has pursued that objective in development of: the eCX; in the Data Sharing Agreement; and in public policy positions
- **Member service objective:** provide precisely defined, reliable clearance for data to drive programmatic interventions against predictable cybercrimes *at any scale required*
- **Operational mission:** archive cybercrime event data into common formats, framed by rigorously prescribed schema; and *curate* them to ensure data fidelity; taxonomic adherence; and auditability / accountability of the data submitters

eCX: Auditable & Provenance Rich

OSINT/*ad hoc* cybersecurity info sources

Publicly available or legally obtainable *information*

Collected from open sources (websites, social media, forums, DNS records, news, blockchains, etc.)

Often noisy, incomplete, and unconfirmed

Lacks guaranteed provenance

Typically point-in-time snapshots

Weak or informal semantics (labels vary by collector)

Useful for discovery and reconnaissance

Evidentiary value often limited

No obligation to correct errors

Examples: scraped phishing URLs, leaked dumps, social media posts

APWG eCrime eXchange data resources

Privately contributed, taxonomically ordered cybercrime-related machine-event data and Internet-event data

Submitted by members from industry, CERTs, multilaterals & LE working under a formal agreement

Curated, normalized, and marked with meaningful, criteria-compliant Confidence Factor indicators

Formal provenance: who submits, under whose authority

22+ year archive of cybercrime event data

eCX's record schema + controlled vocabularies mobilizes a globally recognized cybercrime event-data taxonomy

Used routinely for coordinated mitigation, takedowns, attribution support, forensic narration and historical analyses

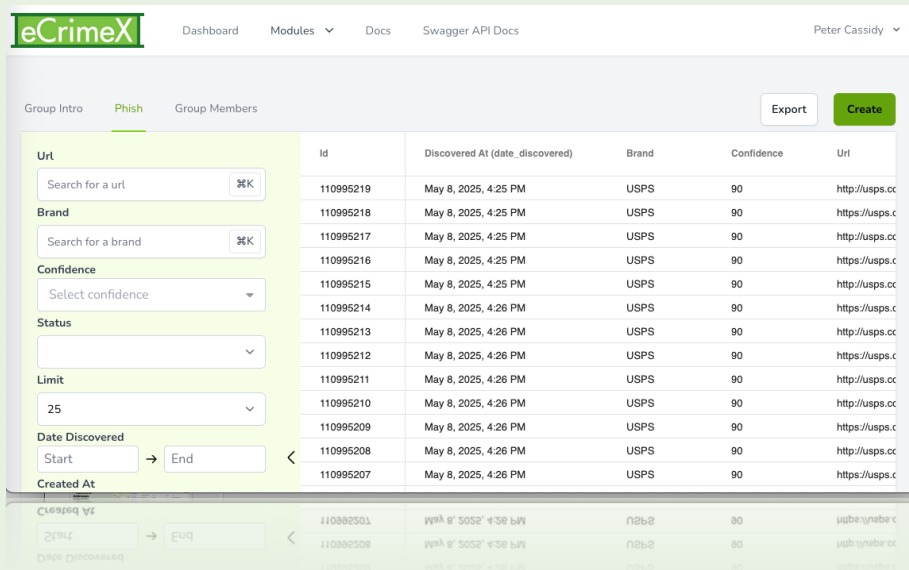
Employed for operational, regulatory, and forensic applications and routinely used for scientific research and experiments

Submitters & members update, retract, & enrich records

Examples: confirmed phishing URLs, confirmed malicious domains, confirmed cryptocurrency cash out wallet IDs

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Since 2004: APWG eCrime eXchange



The screenshot shows the eCrimeX web interface. At the top, there's a navigation bar with 'eCrimeX', 'Dashboard', 'Modules', 'Docs', and 'Swagger API Docs'. Below this, there's a 'Group Intro' section with 'Phish' and 'Group Members' tabs. The 'Phish' tab is active, showing a table of phishing data. The table has columns: 'Id', 'Discovered At (date_discovered)', 'Brand', 'Confidence', and 'Uri'. The data is filtered by 'Brand' (USPS) and 'Confidence' (90). The table lists 10 entries, all discovered on May 8, 2025, at 4:25 PM. The 'Uri' column shows various URLs, mostly from 'http://usps.cc' and 'https://usps.c'.

Id	Discovered At (date_discovered)	Brand	Confidence	Uri
110995219	May 8, 2025, 4:25 PM	USPS	90	http://usps.cc
110995218	May 8, 2025, 4:25 PM	USPS	90	https://usps.c
110995217	May 8, 2025, 4:25 PM	USPS	90	http://usps.cc
110995216	May 8, 2025, 4:25 PM	USPS	90	https://usps.c
110995215	May 8, 2025, 4:25 PM	USPS	90	http://usps.cc
110995214	May 8, 2025, 4:26 PM	USPS	90	https://usps.c
110995213	May 8, 2025, 4:26 PM	USPS	90	http://usps.cc
110995212	May 8, 2025, 4:26 PM	USPS	90	https://usps.c
110995211	May 8, 2025, 4:26 PM	USPS	90	http://usps.cc
110995210	May 8, 2025, 4:26 PM	USPS	90	https://usps.c
110995209	May 8, 2025, 4:26 PM	USPS	90	http://usps.cc
110995208	May 8, 2025, 4:26 PM	USPS	90	https://usps.c
110995207	May 8, 2025, 4:26 PM	USPS	90	http://usps.cc

eCX usage is governed by a Data Sharing Agreement (DSA) amended in 2018 to include a (GDPR-compliant) Code of Conduct for use of the eCX's **/mal_IP** data of malicious IP addresses

- eCX launched as a FTP service in 2004, initially sending phishing URLs from ISPs to anti-virus software developers
- After 2013, eCX relaunched as an API platform with **/endpoints** for **phishing URLs**; malicious **domain names**; malicious IP addresses; and **phishing lure emails** (body and header data)
- By 2024, eCX sourced **cryptocurrency wallet addresses** and **SMS attack records** from members and third parties;
- Data flow consistently moves 2 billion+ data elements monthly. Backplane capacity can support orders of magnitude larger volumes in eCX's current configuration

Key Applications for APWG eCX Data

- Alerting security vendors / browser developers of phishing URLs and IP addresses to update their block lists
- Alerting brand holders & continuity teams about abuses to brands
- Alerting nation states of abuses of key national brands (both sectors)
- Alerting registries & registrars of sponsored domain names abuse
- Scanning ASNs & IP spaces for indications of criminal behavior
- Tracing criminal behavior to specific actors and their histories
- Mapping criminal behavior to IP networks, ASNs, ISPs, Registrars
- Alerting national CERTs of criminality in their ccTLD domain spaces
- Tuning heuristics against sets of confirmed cybercrime event data
- Alerting responders to criminal abuse of Internet infrastructure

eCX Exchange Model Defined

- **eCX Is:** a clearinghouse based on a **centralized** clearance model (versus **bilateral**) in which vetted contributors submit strictly formatted records of *cybercrime-related machine events* and *Internet events*, framed by rigorously delineated categorizations and reliability scores (Confidence Factors)
 - ***Every record with a confidence factor of 90 or 100 is auditable by eCX curators down to the individual submitter . . .***
 - . . . Who can be disciplined and instructed by eCX curators
 - Or removed from the eCX for failure in observing data fidelity requirements set forth in the Data Sharing Agreement
- **eCX is Not:** an open-source repository of similar sorts of data contributed anonymously and pseudonymously and validated in some cases by pseudonymously known parties who may or may not be traceable to liability- and consequence-bearing carbon-based lifeforms

What Kinds of Data Are Shared on the eCX Clearance Platform?

- Known **phishing URLs**
- Full-text **phishing lure emails** with headers
- Malicious **IP Addresses**
- Malicious **domain names**
- Cryptocurrency **wallet addresses** used for cash-outs / ransom payments
- SMS phishing attack **telephone numbers** and related meta-data

[illegible]

Reporting as Keystone to All eCrime Response Routines

- The **/phish** endpoint on the **eCX** archives hundreds of thousands of phishing URLs and meta-data per month from scores and scores of institutions (a large proportion of which are reporting on behalf of any number of enterprises and their brands, trademarks and service marks)

Arizona State University researchers found in their research that **eCX** was alone in its class in this keystone metric: reports lodged at **/phish** on the **eCX** ***generated the highest level of crawler traffic of any entity***^[1]

^[1] **PhishFarm: A Scalable Framework for Measuring the Effectiveness of Evasion Techniques Against Browser Phishing Blacklists** https://docs.apwg.org/documents/phishfarm_ieee_sp_2019_oest.pdf



eCX API Endpoints Review

eCX API Endpoint /phish

- The **/phish** endpoint archives tens of thousands of new, unique **phishing URLs** every month
- The **/phish** records include confidence scores and brand (target) tags, so users can assess record utility for their applications and security routines
- eCX/Phish archives records submitted by APWG members, with confidence factors at 90 or 100

eCX API Endpoint /report_phishing

- The **/report_phishing** endpoint of reported phishing emails, includes the emails' content, including images - and message header data
- Hundreds of thousands of records archived per month
- Records on eCX/report_phishing are submitted by a mix of APWG members (archived at Confidence Factors of 90 or 100; as well as non-member organizations and the general public (archived at Confidence Factor 50)

eCX API Endpoint /mal_ip

- The **/mal_ip** endpoint archives IP addresses that have been recently observed as sources of malicious activity, such as fraudulent transactions, attempted logins and scanning attacks
- Members use the records to manage reputation systems, firewalls, and more
- eCX/Phish archives records submitted by APWG members, with confidence factors at 90 or 100

eCX API Endpoint /mal_domain

- The /**mal_domain** endpoint archives records of suspected and known malicious domain names, including fake stores and fraud/identity-theft websites
- Supporting data can be added and forwarded to registrar and registry operators, or other APWG members, to aid in takedowns or suspensions of abusive domain names
- Records from APWG members, with confidence factors at 90 or 100

eCX API Endpoint /crypto

- The **/crypto** endpoint archives records of problematic virtual currency addresses and helps track suspect transactions, enriching analytic tools' reach and breadth
- Records are useful for cryptocurrency exchanges, wallet providers, trading platforms, and investment funds protecting their customers against phishing and cybercrime
- Some records with high confidence factors from APWG members
 - most, from 3rd parties, archived at Confidence Factor 50

eCX API Endpoint /malicious-sms

- The /**malicious-sms** endpoint archives records of phishing attacks made through SMS and text communications media
- Malicious-sms endpoint records are useful for: blocking lists serving users; hygienic routines for network management; and forensic applications for operations and law enforcement personnel
- Some records are contributed by APWG members with Confidence Factor at 90 or 100, most others, 3rd party reports with CF 50

Report Confidence Scoring

- All of the eCX API /endpoints require a record to have a **Confidence Factor** that relates to the degree of manual, human processing used in scoring the report as an indicator of the its reliability. As follows:
- **100 CF** = Manual inspection/confirmation by an APWG member reporter, submitting data under terms set out in the APWG Data Sharing Agreement
- **90 CF** = Automated inspection/confirmation by Artificial Intelligence or Machine Learning systems directly curated by an APWG member submitter, under terms set out in the APWG Data Sharing Agreement
- **50 CF** = Machine data-capture or third-party volunteered report ***without*** confirmation by APWG member personnel or by member-curated AI/ML
 - Third-party source or general public sourcing so - on its face, without further meta-data that would speak to reliability. The assumption is that the record has a 50/50 chance of being accurate
- **0 CF** = False positive

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Data Programs Documentation

See: APWG eCX Github Page:

<https://github.com/APWG/ecx>

eCX API Documentation:

<https://apwg.github.io/ecx2-openapi-doc/#/>

[eCX Resources / Trust Architecture Discussions](#)

<https://apwg.org/ecx/>



Sectoral Risk Data Exchange for Cybercrime Events

Since 2004

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Why an *NGO-Managed* Clearinghouse?

- When *everyone* (businesses, LEOs, churches and taverns) fights *the same threats* (fires; pirates; & measles), civilizations forge *common defense* infrastructures
- For the above threats, non-profit clearinghouses of fire histories (underwriter co-ops); piracy maps and events (maritime organizations); & case data (public health/vaccination management) are maintained to manage predictable risks
- In today's global confrontation against cybercrime:

APWG eCrime eXchange (eCX) – *since 2004*

- For equitable balance of risk management, access to data, and relevance of data corpora, an independent, user-supported NGO model *just works* – as it always has – for as long as humanity has been managing predictable risks

Sectoral Risk Exchanges Examined

Instrumentation	APWG eCrime eXchange	Property & Casualty Clearinghouses	DNA Sequence Archives	Maritime Piracy Clearinghouses
Data model type	Typed event-based model (phishing, IPs, etc)	Form-based, loss-event & actuarial data schemas (e.g. claims data)	Biological/genetic ID + metadata (e.g. DNA sequences)	Typed event-based model (piracy / armed; robbery; time; lat/long; vessel)
Controlled vocabularies	Yes — threat types, brand targets, entity tags	Yes — cause of loss, policy class, geographic codes	Yes — <i>clade</i> names, hosts, variants, mutation codes	Yes — ship types, attack method, region, weapons
Provenance tracking	Yes — high-confidence reports tied to pre-registered entities	Yes — insurer ID and state regulator traceability	Yes — submitter name, lab, affiliation, country	Yes — vessel, flag state, reporting authority, IMO number
Immutable submission identity	Yes — each submission ID linked to DSA signer	Yes — required by trade organization rules & state legal statutes	Yes — (e.g. GISAID DAA), mandatory data field	Yes — incident / case ID logged in IMO (UN) / IMB (ICC) system
Confidence/quality score	Yes — per record, machine-readable	Indirect — via actuarial / statistical reliability	Partial — sequence quality scores + metadata checks	Indirect — factual reports, later verification; no machine-readable score
Change tracking/versioning	Yes — all record edits logged, authority restricted	Yes — versioned filings, statutory audit trails	Yes — updates logged, sequence revisions traceable	Yes — updates when more details confirmed; revisions logged
Legal framework governing data semantics	Yes — APWG Data Sharing Agreement (DSA) governs meaning	Yes — state/federal insurance code governs data meaning	Yes — (e.g. GISAID Data Access Agreement [DAA] enforces data use & terminology)	Yes — Maritime Law & IMO conventions (reporting obligations, ICC IMB practice)

What do you get once data is exchanged programmatically?

- Archive fusion
- Power: More event data to fuel prevention, response & forensics applications as broadly as possible
- Velocity: Faster event data to speed response & prevention systems as quickly as possible
- Profundity: deeper views and broader forensic narrative-building capacity
- Collaborative flexibility
- Triumph: (or at least reasonable control) over predictable risks – eventually – as ***programmatic*** responses drive those expressed risks to marginal significance

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eCX Data Correspondents' Trust Infrastructure

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APWG Data Mobilization Strategy: Cultivate Data Sharers' Trust

- Data sharing – particularly for *impeding* events – is all about trust
- Everybody following explicit and precise usage rules builds trust
- Articulation of common expectations and action executing them in deed builds trust
- Empowerment as a data contributor as well as consumer builds trust

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Data Sharing Agreement: Key to Cultivation of Data Sharer Trust

- APWG treats member / data-sharers operating under our DSA like equals
- APWG members under DSA have full write access to the eCX API endpoints – exactly the same as members of the APWG board of directors
 - New reports and ability to make corrections to existing reports
- DSA spells out explicitly: data usage/license terms; submission rules; and liability/indemnification scheme
- **Corporate counsel has what he or she needs to determine if risk attendant data exchange has been reasonably addressed on behalf of their enterprises**
 - Without that, for incorporated entities of any size, programmatic data exchange cannot be animated

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Auditability Key to Trust – Like Any Business Process

- APWG eCX does not release the identity of the data submitter but data submission is **NOT** anonymous
- **By design**, submitters **can be – and are, routinely – traced** to submitted data, if and when the APWG's administrative personnel need to resolve data fidelity issues or operational exigencies

Terms of Usage Bounds Trust Domain and Enables Accountability

- Data submitters need assurance clearinghouse members can't do silly things with their data
 - Data may be kind-of sensitive (i.e. "we found it before other people did") and I may not want my sensitive data going public / rogue
- When member institutions submit data, management and operations want to know what other members can do with it
- The DSA frames and answers those questions so corporate counsel can make a rational decision on participation
- Data flows only after counsel decides affirmatively

Data Sharing Agreement + Auditability Balances Interests

- APWG works with operations personnel most often – but our DSA recognizes that *corporate counsel* directs participation
- Operations gets its data – and counsel through the DSA receives instrumentation essential to their risk management mission
- And APWG retains authority to act on behalf of users in maintaining: data fidelity; taxonomic coherence; and timeliness
- With the DSA and the neutral curator, data sharers are assured:
 - A) You know the rules;
 - B) Other data sharers know the rules, too, and;
 - C) APWG, the curator, can see if you follow the rules or not - discipline data correspondents participating in the clearinghouse

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Contact: APWG Engineering

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*Global forge of counter-cybercrime
data logistics and policy since 2004*